NUMERICAL DESIGN OF DUAL-BELL NOZZLE

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SOMMARIO

La risoluzione dei problemi inversi riguarda la determinazione della forma geometrica delle pareti di un corpo che deve soddisfare alcune proprietà fluidodinamiche prescritte su di esse. Il punto di partenza di questa metodologia è basata su una tecnica "time-dependent" che si è dimostrata molto accurata nella risoluzione dei problemi inversi applicati ai flussi euleriani. Lo scopo del presente lavoro è quello di estendere il metodo inverso, per il progetto di ugelli assialsimmetrici in flussi rotazionali e viscosi. In particolare, la procedura numerica verrà utilizzata per il progetto di un ugello a doppia campana "Dual-Bell". I risultati numerici saranno quindi validati con dati sperimentali disponibili in bibliografia.

ABSTRACT

Inverse problems consist in finding the shape that the walls of a body have to assume to satisfy some flow properties prescribed over them as design data. Based on earlier numerical work published by the authors, different advance nozzle configurations will be designed and tested. The starting point of the approach is a time-dependent technique which has shown to be quite effective in solving numerically inverse problems under the assumption of inviscid, rotational flow fields. The aim of the present work is the developing of a newly improved method to solve inverse problems for viscous, compressible, rotational flows for axysimmetric design of Dual-Bell nozzles. The numerical procedure will be validate against experimental data.

1. INTRODUCTION

The dual-bell nozzle concept, first proposed by Forster and Cowles[1] in 1949, has recently gained renewed interest in the Unites States, Japan, Europe and Russia. The reason for this interest is the unique feature of an one-step altitude adaptation, achieved only by a wall inflection and, thus, without moving parts. Because of the classical bell-shaped form, necessary changes on the launcher level are small when compared to other altitude adaptive nozzle concepts, for example plug nozzle. At low altitudes, controlled and symmetrical flow separation occurs at this wall inflection (see Fig 1a), which results in a smaller effective area ratio without generating dangerous side loads. At higher altitudes, the nozzle flow is attached to the wall until the exit plane and the full ratio is used (Fig 1b). Because of the higher area ratio, an increase in vacuum performance is achieved. The first published experiments on dual-bell nozzle proved the altitude adaptive capability of this nozzle, but also showed that additional losses are induced compared to a conventional bell-shape design. studies. The dependence of additional losses induced in dual-bell nozzles as a function of nozzle designed parameters was systematically investigated, and performance characteristics were compared to a conventional reference bell nozzle. It was shown that additional losses induced by the wall inflection depend on the overall contour design and can be surprisingly low. In 1999, Frey and Hagemann[3] presented results of analytical and numerical studies. Figure 2 shows, dual-bell performance characteristic vs flight altitude. At sea-level operation, the pressure within the separated flow region of the dual-bell nozzle extension is slightly below the ambient pressure, inducing a thrust loss called aspiration drag. In addition, the flow transition from sea-level mode to altitude mode occurs before the optimum crossover point, which leads to a thrust loss compared to an ideal switch-over. The non-optimum contour of the full-flowing dual-bell nozzle in the altitude mode results in further losses.

This paper is a logical continuation of the work developed by Ferlauto and Marsilio[2] in 2002. In Ref.[2] a numerical inverse methodology for the compressible Navier-Stokes equations was developed and validated for 2-D geometry. Inverse problems consist in finding the shape that the walls of a body have to assume to satisfy some flow properties prescribed over them as design data. A classical problem is to find the wing profile which realizes a given pressure distribution. First, an extension for axysimmetric geometry is implemented. After that, few design geometries carried out by inverse method design for dual-bell nozzles will be presented and validated against experimental data. The design considerations are also discussed.

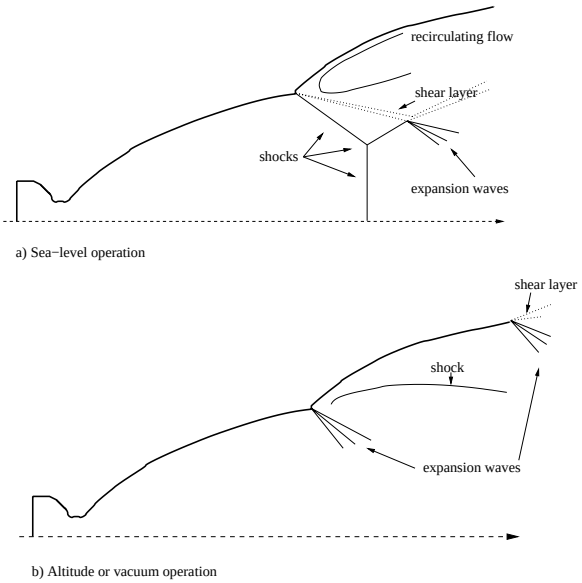


Figure 1: Different flow regimes of a dual-bell nozzle

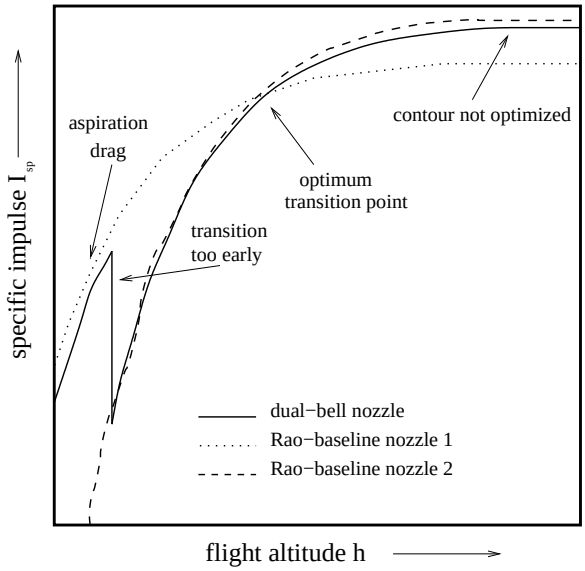


Figure 2: Performance characteristic of a dual-bell nozzle

2. DESIGN ASPECTS OF DUAL-BELL NOZZLES

Principle design considerations for dual bell nozzles have been discussed in detail in Ref.3. In the following subsections, brief summary is given.

Inner Contour, Base Nozzle. The inner contour par of dual-bell nozzles, also name base nozzle, may be designed as a Rao-type bell nozzle for maximum performance or as a truncated ideal nozzle. Recent investigations performed in Europe within the frame of flow separation control device program have shown that parabolic or thrust-optimized nozzles feature a specific plume pattern, the cap-shock pattern.[4] This plume pattern may trigger a transition from free to restricted shock separation. This potential occurrence of restricted shock separation in the base nozzle of a dual-bell nozzle has to be taken into account within the design process.

Wall Inflection and Nozzle Extension. The wall pressure profile of a dual bell nozzle within the base nozzle is equivalent to that of conventional bell nozzles. At the wall inflection, an expansion fan accelerates the flow. To ensure controlled flow separation at the wall inflection within a certain range of ambient pressure, the required wall pressure value behind the wall inflection is estimated with commonly used separation criteria.[3]

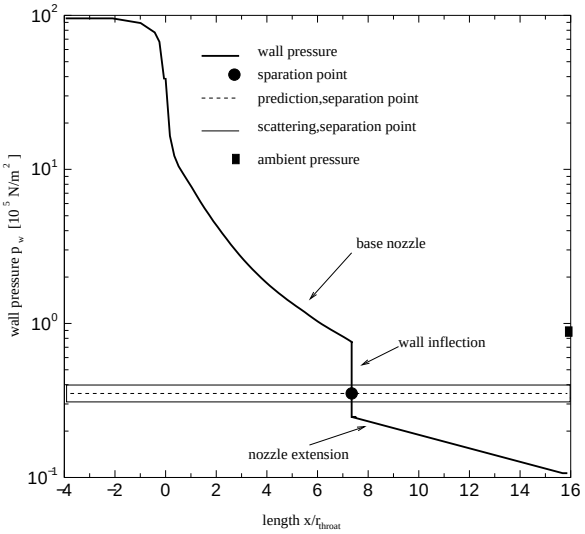


Figure 3: Wall pressure profile for expansion into vacuum

The corresponding wall inflection angle is calculated using the well-known Prandtl-Meyer equations.

In a graph showing wall pressure vs nozzle length , as shown in Fig. 3 for a dual-bell nozzle configuration described in detail in Ref. 3, the predicted separation pressure $p_{w.sep}$ is introduced as a limiting line. The nozzle part with wall pressure value above the line is full flowing, whereas the region with pressure values below is fully separated. During the sea-level mode, the pressure limit given by the line of the separation criterion must, therefore, intersect the wall pressure profile within the pressure drop at the wall inflection, as shown in Fig. 3. This controlled flow separation at the wall inflection prevents the generation of dangerous side loads commonly observed in conventional overexpanded nozzles featuring uncontrolled flow separation.

Ambient pressure continuously decreases during the ascent of the rocket. In the wall pressure graph shown in Fig.3, this is linked to a decrease of the separation pressure and, thus, to a movement of the separation line toward lower pressures. At a certain ambient pressure, the bar of the separation criterion intersects the wall pressure profile in the nozzle extension. Different designs of this extension, producing a negative, a zero, or even a positive wall pressure gradient, will lead to different flow behavior.

- Nozzle extension with negative (or favorable) wall pressure gradient leads to an uncontrolled flow separation within the nozzle extension, as can be observed in conventional conical or bell-shaped nozzles.
- Within a nozzle extension with zero wall pressure gradient (constant pressure extension), the horizontal bar characterizing the predicted flow separation point reaches all points of the nozzle wall at the same time. This might lead, at least theoretically, to a quick change from sea-level mode to altitude mode. However, in principle, strong nonsymmetric separation, or even a pulsation of the separation point, could be expected fro a theoretical point of view.
- For nozzle extension with positive (or adverse) wall pressure gradient (overturned extension), the predicted separation point intersects the wall pressure profile first in the exit plane of the nozzle, but the full flowing of the nozzle extension first occurs at ambient pressures, where the bar reaches the lowest wall pressure at the wall inflection. Therefore, this type of nozzle expansion also might lead, at least theoretically, to a quick change from sea-level mode to altitude mode.

3. MATHEMATICAL MODEL

We formulate an inverse problem in which the location of the contact interfaces is not known a priori and their final shape is part of the solution. At the same time other design requirements, expressed in terms of pressure distributions, can be prescribed to obtain the desired nozzle performances. The design problem of finding the shapes of intakes or nozzles that satisfy the prescribed pressure distributions is solved here according to

a time-dependent process.[5] Interfaces separating internal flows are fitted, as in the jet of multiple nozzles and the external flow, while shocks or contact discontinuities inside each flow are inherently captured by the numerical scheme. From the numerical point of view the method is based on the finite volume projection of the time-dependent Euler equations according to a second order accurate ENO scheme.[6]

3.1 GOVERNING EQUATIONS

Compressible viscous flows are governed by the Navier-Stokes equations.All this set of equations may be written in a compact integral conservative form as:

$$\frac{\partial}{\partial t}\int\limits_{v}\vec{W}dv+\int\limits_{s}\vec{F}_I\cdot\vec{n}ds+\int\limits_{s}\vec{F}_V\cdot\vec{n}ds=0 \tag{1}$$

where v represents an arbitrary volume enclosed in a surface s. System (1) can be reduced to non-dimensional form with the help of the following reference values: L for length, ρ_∞ for density, T_∞ for temperature, $\sqrt{RT_\infty}$ for velocity, RT_∞ for energy per unit mass and μ_∞ for viscosity. Therefore, from now the flowfield variables should be considered as non-dimensional. In particular, $\vec{W}$ is the hyper-vector of conservative variables, tensor $\vec{F}_I$ contains the inviscid fluxes, tensor $\vec{F}_V$ contains the viscous fluxes

$$\begin{aligned}\vec{W}&=\{\rho,\rho\vec{q},E\}^T\\ \vec{F}_I&=\left\{\rho\vec{q},p\bar{\bar{I}}+\rho\vec{q}\otimes\vec{q},(E+p)\,\vec{q}\right\}^T\\ \vec{F}_V&=\frac{\sqrt{\gamma}M_\infty}{\mathrm{Re}_\infty}\left\{\vec{0},-\bar{\bar{\tau}},-K\nabla T-\bar{\bar{\tau}}\cdot\vec{q}\right\}^T\end{aligned} \tag{2}$$

Quantities ρ, p and $\vec{q}=\{u,v,w\}^T$ are the local density, pressure and velocity, respectively; E represents the total energy per unit volume:

$$E=\rho\left(e+\frac{q^2}{2}\right) \tag{3}$$

where e is the internal energy per unit mass, M_∞ and Re_∞ are the freestream Mach number and the Reynolds number, γ is the ratio of the specific heats, $\tilde{\nu}_T$ is the modified eddy viscosity and finally $\bar{\bar{I}}$ is the unit matrix. The viscous stresses $\bar{\bar{\tau}}$ are contained in tensor , given by:

$$\tau_{ij}=\mu\left[\frac{\partial q_j}{\partial x_i}+\frac{\partial q_i}{\partial x_j}-\frac{2}{3}(\nabla\cdot\vec{q})\delta_{ij}\right] \tag{4}$$

The thermal conductivity K is calculated in non-dimensional form as

$$K=\frac{\gamma}{\gamma-1}\left(\frac{\mu}{Pr}\right) \tag{5}$$

where Pr is the Prandtl number. The viscosity μ is computed via Sutherland's law as

$$\mu=T^{3/2}\left(\frac{1+T_{ref}}{T+T_{ref}}\right),\quad T_{ref}=\frac{110.4}{T_\infty} \tag{6}$$

Finally, the perfect gas relationship , $p=\rho T$, completes the set of equations. Eqs. (1) are approximated by a finite volume technique by discretizing the (x,y) plane by means of four sided cells whose shape depends on time. The integration in time is carried on according to a Godunov type two-step scheme[7] . At the *predictor* step, a standard first order FDS is used: the primitive variables (ρ, p, e, u, v) are assumed as an averaged, constant value inside each cell. The fluxes $\vec{F}_I$, are evaluated by solving the Riemann's problems pertinent to the discontinuities that take place at the cells interfaces. To this purpose, we adopted the approximate Riemann solver suggested by Pandolfi.[8] At the *corrector* level, the second order of accuracy is achieved by assuming a linear, instead of constant, behavior of the primitive variables inside the cells, according to the ENO concept.[6] The resulting scheme is second order accurate in both time and space.[7] Velocity and temperature gradients needed to evaluate viscous fluxes[9] $\vec{F}_V$ in corrispondance with lateral surface are computed through a standard technique that uses central differences and applies the Gauss's Theorem.

3.2 AXIS-SYMMETRIC FORMULATION

System (1) is integrated in time starting from an arbitrary initial solution until steady-state conditions are reached, in the spirit of a time-marching technique. Governing equations are discretized by means of a finite volumes approximation. In case of an axis symmetric flow, system (1) can be reduced to a two-dimensional form, where the two independent variables are the axial direction x and the radial direction y. The discretized system, using an explicit integration method, takes the following form:

$$\mathbf{W}_{I,J}^{k+1} = \mathbf{W}_{I,J}^{k} - \frac{\Delta t}{A y_c}\left[\sum_{i=1}^{4}(\mathbf{F_I}_i + \mathbf{F_V}_i)s_i y_i - \mathbf{H_I} - \mathbf{H_V}\right] \tag{7}$$

where

$$\mathbf{H_I} = \{0, 0, pA, 0\}^T, \qquad \mathbf{H_V} = \left\{0, 0, -\frac{\sqrt{\gamma}M_\infty}{Re_\infty}\tau_{\theta\theta}A, 0\right\}^T \tag{8}$$

with

$$\tau_{\theta\theta} = \mu\left[\frac{4}{3}\frac{v}{y_c} - \frac{2}{3}(u_x + v_y)\right] \tag{9}$$

Source terms $\mathbf{H_I}$ and $\mathbf{H_V}$ account for the effect of pressure and shear stress on lateral surfaces of the elementary volume, respectively. Viscous stresses contained in $\mathbf{F_V}$, when written in cylindrical coordinates under the assumption of axially symmetric flow, read like:

$$\tau_{xx} = \mu\left[\frac{4}{3}u_x - \frac{2}{3}\left(v_y + \frac{v}{y}\right)\right] \tag{10a}$$

$$\tau_{xy} = \mu(u_y + v_x) \tag{10b}$$

$$\tau_{yy} = \mu\left[\frac{4}{3}v_y - \frac{2}{3}\left(u_x + \frac{v}{y}\right)\right] \tag{10c}$$

The two-dimensional representation of an elementary control volume is shown in Fig. 4. Symbol A represents the cell surface, s_i are its sides, y_i are the average distances of the sides from the symmetry axis and y_c is the distance of the cell baricenter from the symmetry axis.

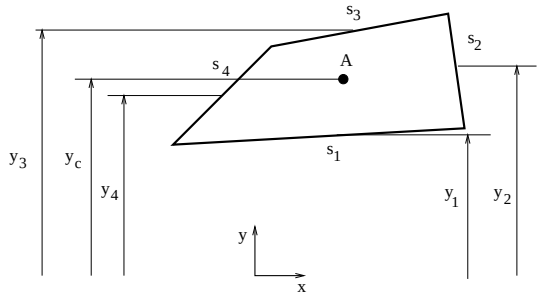


Figure 4: The finite volume in the axially symmetric case

3.3 BOUNDARY CONDITIONS

The computational domain is bounded by artificial (i.e. far field boundaries) and physical contours (i.e. impermeable walls), which can be solid, as in the direct problem, flexible, as in the inverse problem, or partly solid and partly flexible. The theory of characteristics states the number of conditions to be imposed at each boundary of the computational domain. One boundary condition is needed at impermeable boundaries as it is prompted by the physics of the problem. At solid walls the vanishing of the velocity ($\vec{q} = \vec{0}$) to the wall is imposed, while at a flexible wall, according to the inverse problem formulation, the design pressure is prescribed. The computation at the boundaries is carried out by solving an *half Riemann problem*. At deformable walls, once the flow properties are computed, the geometry is updated by enforcing the condition of impermeability: each point of the wall has to move with a normal velocity equal to the normal component of the flow velocity, as described with details in Zannetti.[5]

The same method here described to solve inverse problems can be used to determine the shape of plumes and interfaces. For instance, the stream-tube confining a jet in air at rest can be seen as the wall of a duct along which a constant pressure is prescribed. Moreover, in flow regions such as after-bodies or dual nozzles in by-pass turbofan, contact discontinuities are generated by the different stagnation conditions and thermodynamic properties of the incoming flows. Such discontinuities are interfaces that can be computed explicitly according to the present method: they are considered as impermeable and deformable walls across which pressure and normal component of the flow velocity are imposed to be continuous.[2]

4. NUMERICAL RESULTS

For the numerical verification of the aforementioned inverse problem technique three different truncated base nozzles were designed and compared with the experimental data reported in Ref. 10. The overall design parameters were kept constant for all dual-bell nozzles and are summarized in Table 1.

Design criterion	Value
Driving gas	Nitrogen
Pressure ratio for full-flowing of base nozzle	$p_c/p_a > 50$
Pressure ratio for fixed separation at wall inflection	$50 < p_c/p_a < 100$
Pressure ratio for full-flowing of nozzle extension	$p_c/p_a > 200$
Area ratio of base nozzle	$\tilde{\epsilon} = 16$
Area ratios of nozzle extensions	$\epsilon = 35 - 45$
Nozzle throat radius	$r_{throat} = 10$ mm
Length of dual-bell nozzle	$x/r_{throat} = 17$

The prescribed wall pressures imposed for the numerical design are shown in Fig.5. Figure 5 shows the three

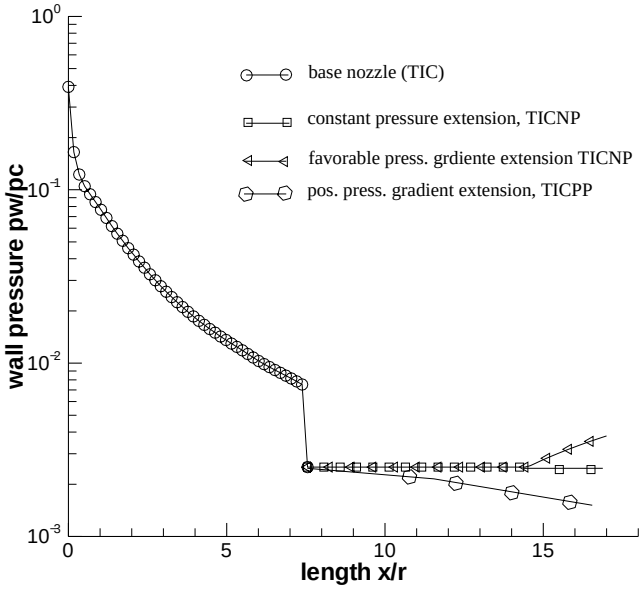


Figure 5: Prescribed wall pressure profiles for cold-gas dual-bell nozzle

pressure distributions for the truncated ideal base nozzle: truncated ideal base nozzle and nozzle extension with favorable wall pressure gradient TICNP, truncated ideal base nozzle and a nozzle extension with zero wall pressure gradient TICCP, and truncated ideal base nozzle and a nozzle extension with positive wall pressure gradient TICPP. The impose pressure distributions is then enforced as a target pressure at the dual-bell nozzle and the inverse problem is solved for the nozzle geometry. No matter what the initial condition are, we expect the final geometry to accurately replicate the target nozzle geometry. The numerical tests was solved on a 100×52 grid. Stretching was used in the y-direction (the radial one), to better refine the region close to the upper and lower boundaries. The bottom boundary is implemented as a axis-symmetric conditions, while the upper boundary moves according to the procedure of inverse designing. A sequence of the flowfield evolution during the transient is presented in Figure 6. The starting geometry is visible in Figure 6a. The initial flowfield

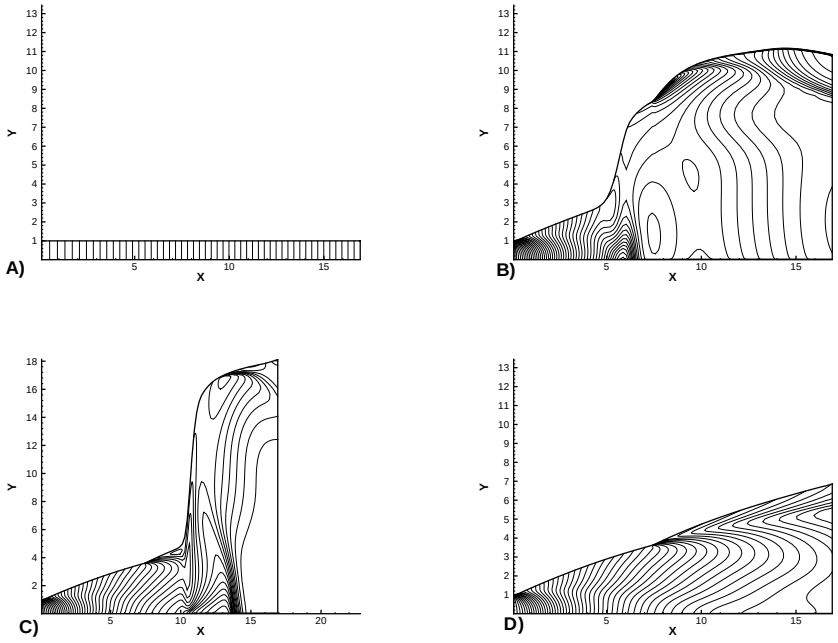


Figure 6: Flowfield evolution: A) initial; B) C) transient D); final configuration

can be considered sufficiently far from the target one. The inverse problem procedure is in fact driven by flow evolution, so that the complexity of the initial flowfield rather that the initial geometry of the domain, is of relevance. The nozzle shape is obtained with high accuracy. The target and the computed geometry are indistinguishable. Figure 7 shows the different bell shapes obtained for the TICNP, TICCP and TICNP nozzles. The Mach number contour lines for the TICNP nozzle are shown in Fig. 8.

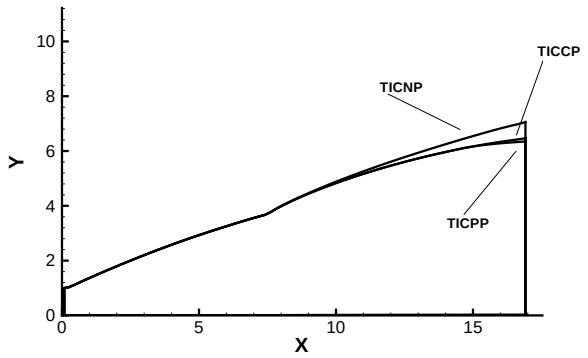


Figure 7: Final Geometries for TICNP, TICCP, TICPP

Wall pressure data p_w/p_c computed by the inverse method compared with the experimental results are shown in Figure 9. Pressure data are shown for $30 < p_c/p_a < 50$, where p_c and p_a are the chamber and the ambient pressure, respectively. Also reported is the comparison between the experimental designed pressure (**epf**) with the numerical computed (**cpf**). The results confirm the good agreement between the computed numerical data and the experimentals.

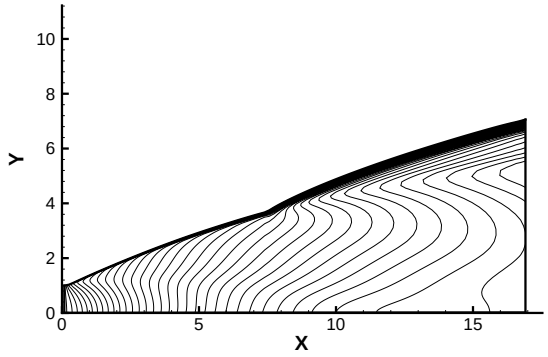


Figure 8: Mach Number contour lines for TICNP

5. CONCLUSIONS

An approach to the numerical solution of an inverse problem using Navier-Stokes equations to design Dual-Bell nozzles has been proposed and validate. The tests have shown that the technique gives an accurate representation of both target flowfield geometry and of the flow solution. The computational cost is that of an unsteady direct flow simulation. Since only the steady configuration is of interest, the procedure coul be greatly enhanced using over-relaxing techniques. The extension to three-dimensional flows is straightforward, at cost of an unstaedy flowfield evaluation.

REFERENCES

[1] C. Foster and F. Cowles, "Experimental study of gas-flow separation in overexpanded exhaust nozzles for rocket motors." Jet Propulsion Lab. Rept.4-103, 1949.

[2] M. Ferlauto and R. Marsilio, "Inverse problem solution using navier-stokes equations." DIASP Report No. , submitted for publication on AIAA Journal, 2003.

[3] M. Frey and G. Hagemann, "Critical assessment of dual-bell nozzles," *Journal of Propulsion and Power*, vol. 15, pp. 137–143, 1999.

[4] M. Frey and G. Hagemann, "Restricted shock separation in rocket nozzles," *Journal of Propulsion and Power*, vol. 16, pp. 478–484, 2000.

[5] L. Zannetti, "Time dependent method to solve inverse problems for internal flows," *AIAA Journal*, vol. 18, pp. 754–758, 1980.

[6] A. Harten, B. Enquist, S. Osher, and S. Chakravarthy, "Uniformly high oder accurate essentially non-oscillatory schemes iii," *J. of Computational Physics*, vol. 71, 1987.

[7] A. D. Mascio and B. Favini, "A two step godunov-type scheme for the euler equations," *Meccanica*, vol. 26, pp. 179–189, 1991.

[8] M. Pandolfi, "A contribution to the numerical prediction of unsteady flows," *AIAA Journal*, vol. 22, pp. 37–46, 1983.

[9] R. . Marsilio, "Numerical simulations for plug nozzle flowfields." AIAA Paper 2001-0670, 2001.

[10] G. Hagemann, M. Terhardt, and D. Haeseler, "Experimental and analytical design verification of dual-bell concept," *Journal of Propulsion and Power*, vol. 18, pp. 116–122, 2002.

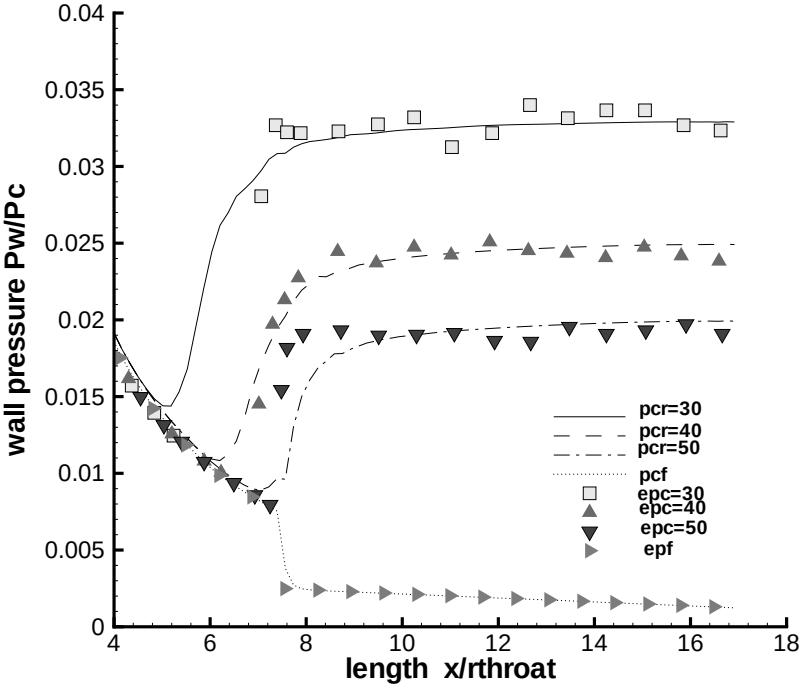


Figure 9: Wall pressure comparison